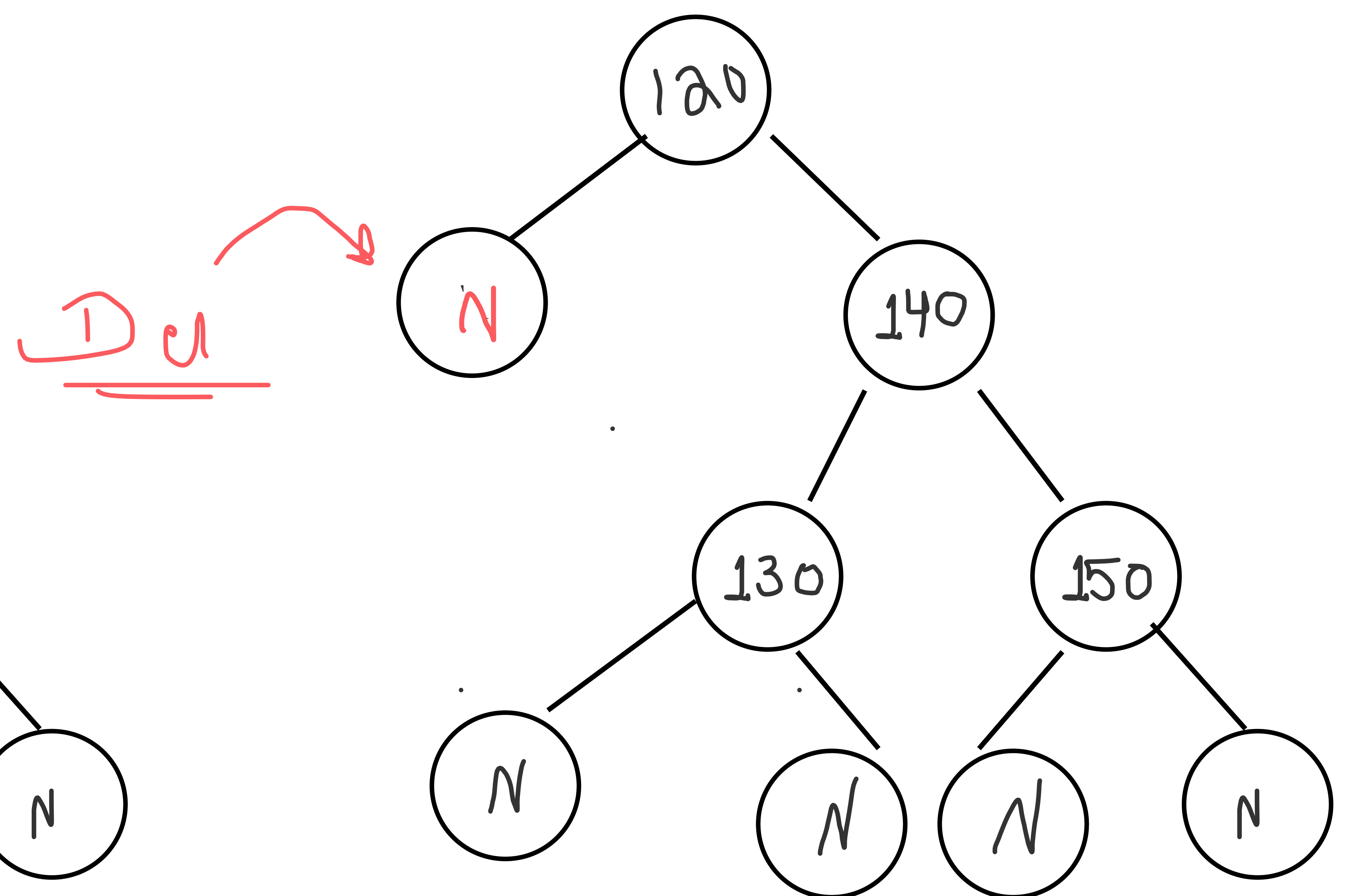
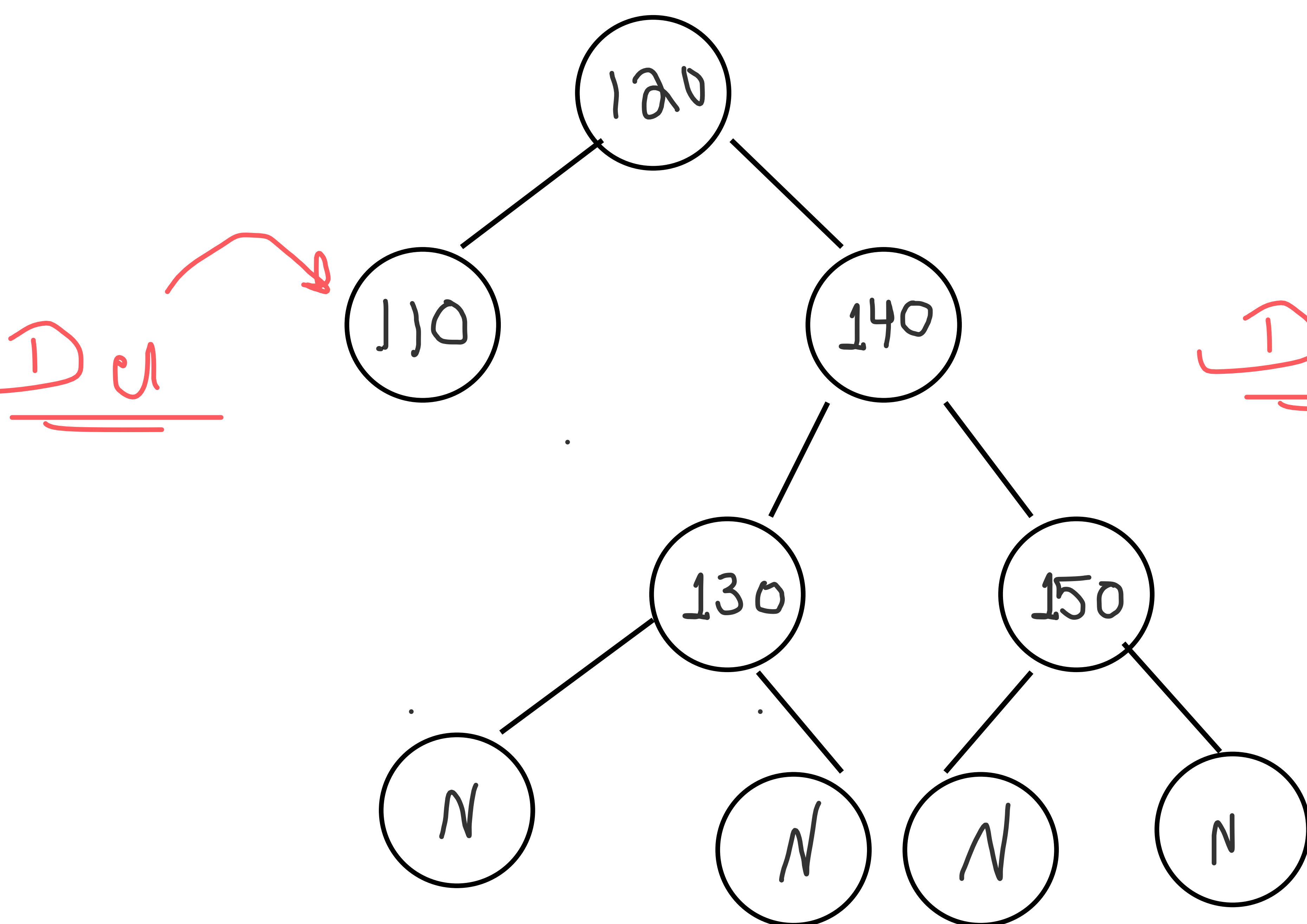
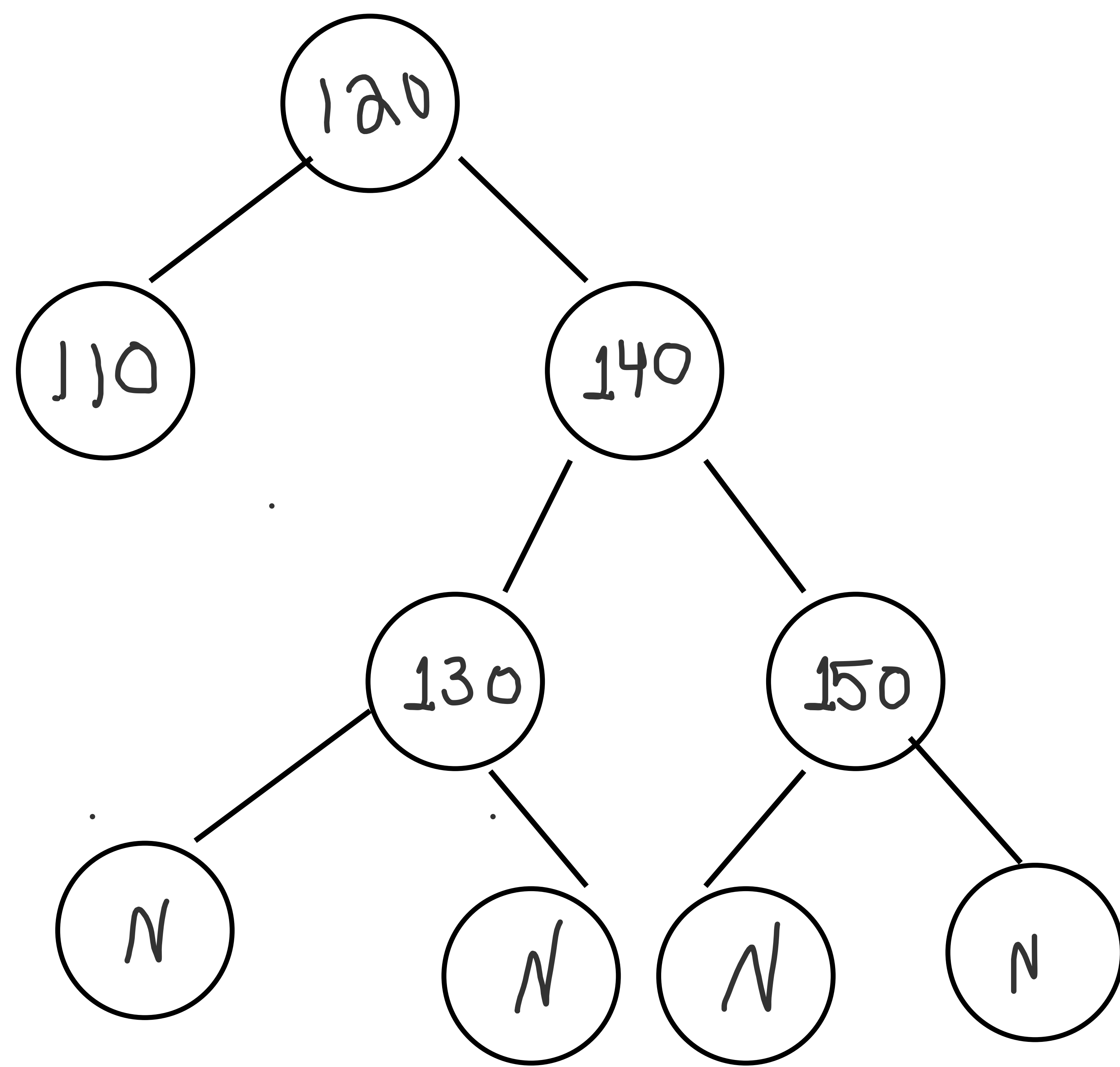
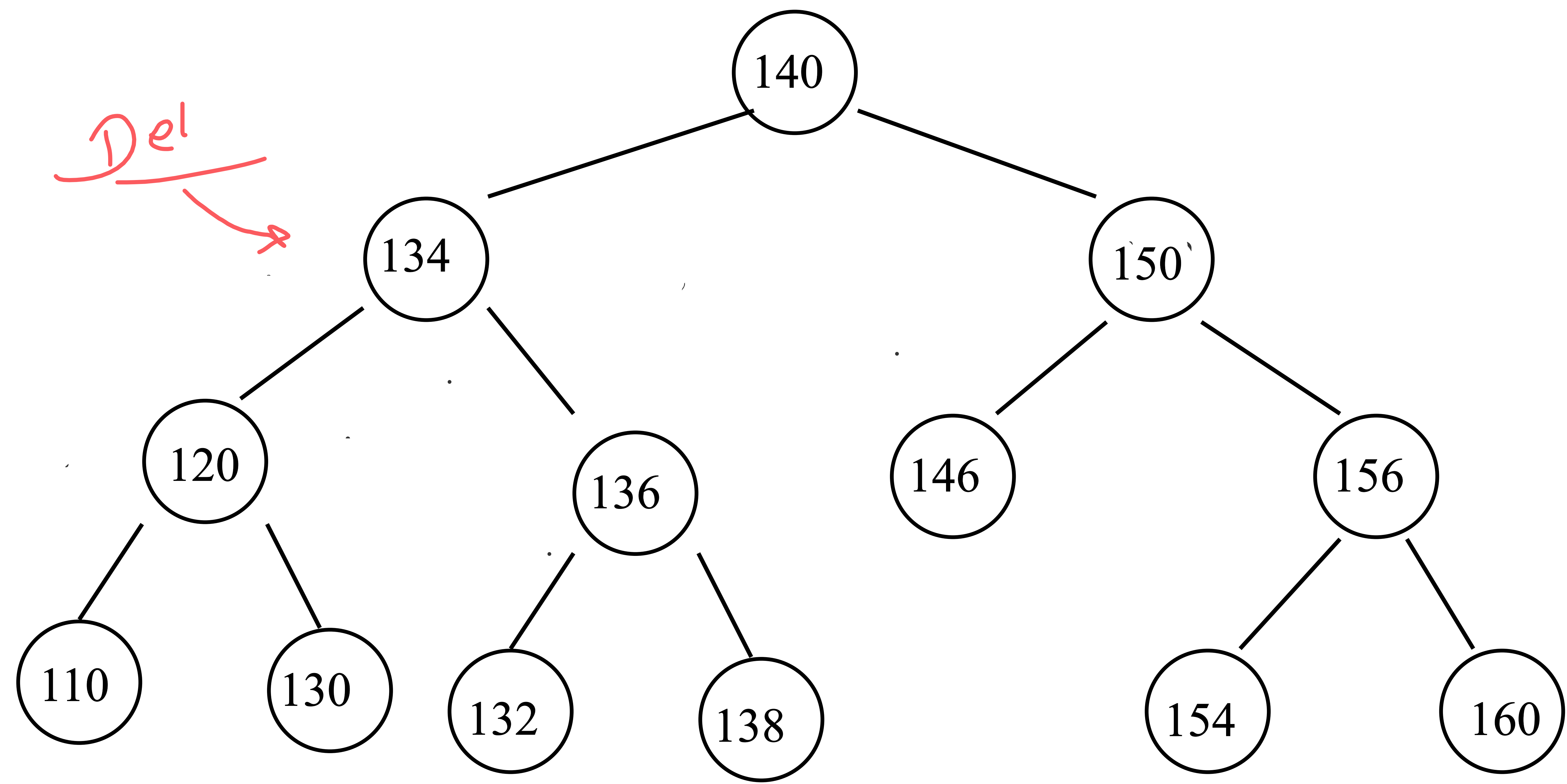


AVL tree Delete



for cond 4d

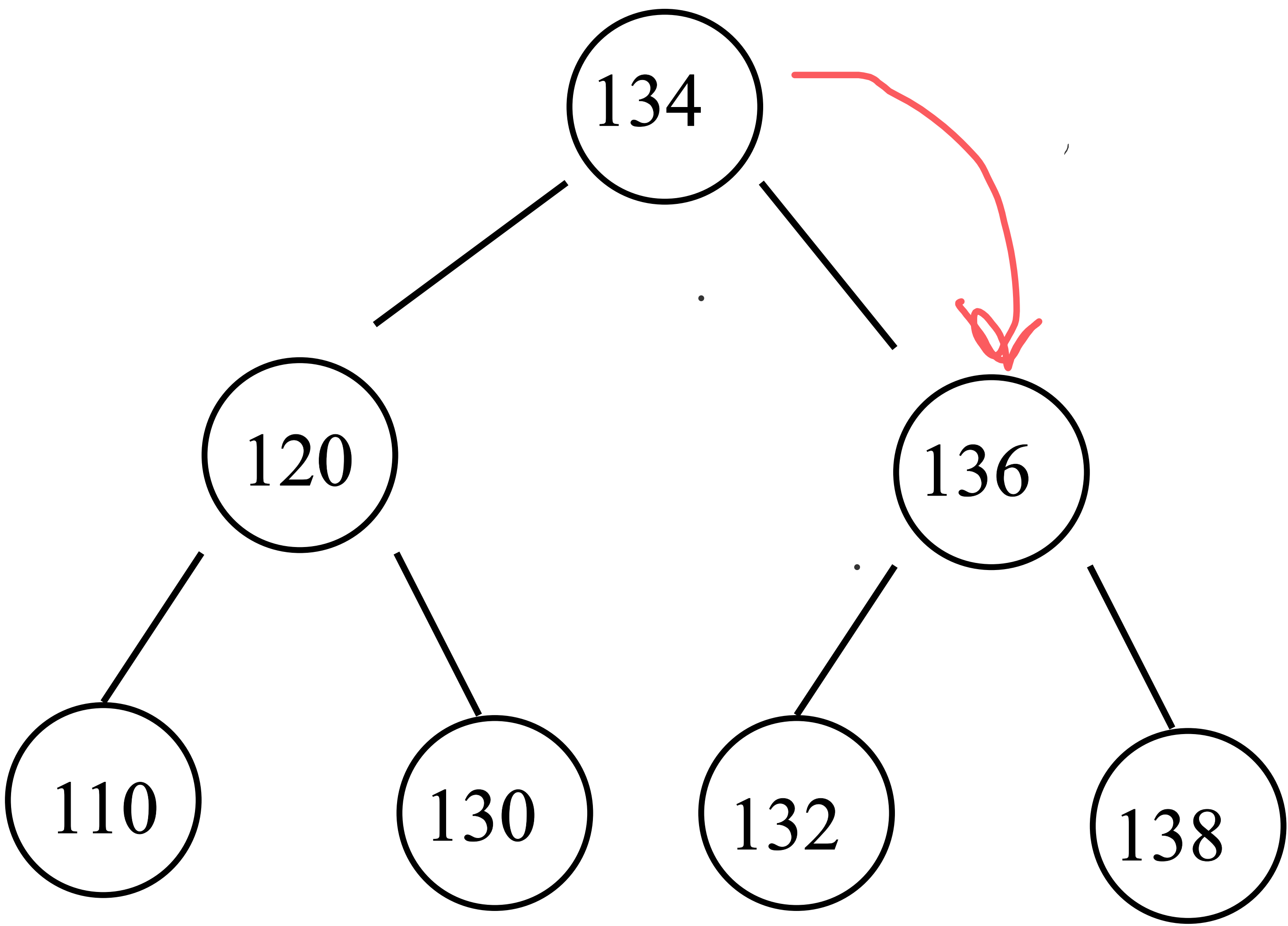
for Simple tree for CLARITY



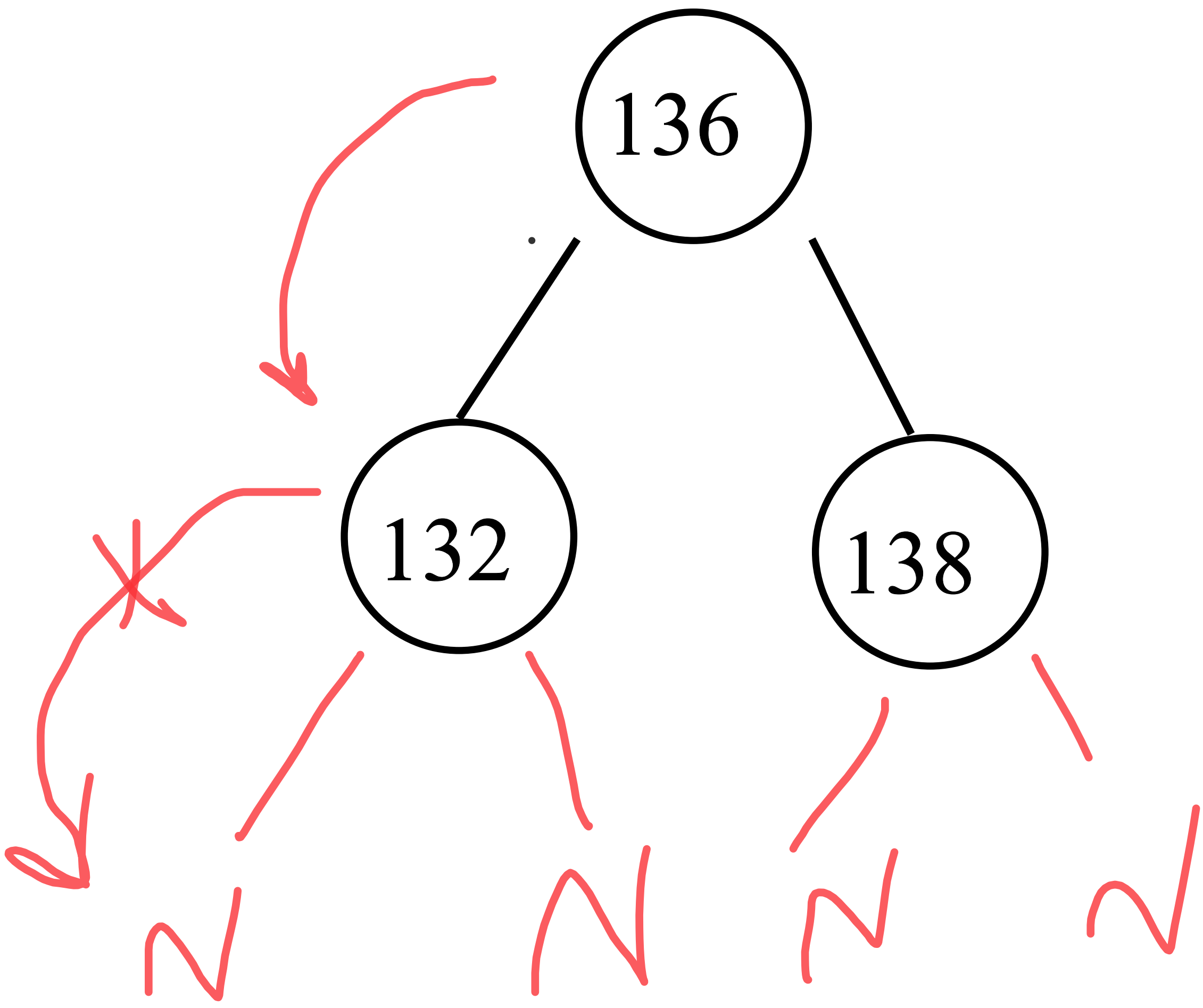
line 251

temp = mini (root-> right)

root->right = 136



tree* mini(tree*root)
{
 while(root->left!=NULL)
 {
 root=root->left; 132
 }
 return root; → 132
}



`temp=mini(root->right);` → 132

`root->data=temp->data`

→ 134 = 132

root updated here from 134 to 132

`root->right = Delete (root -> right , temp -> data)`

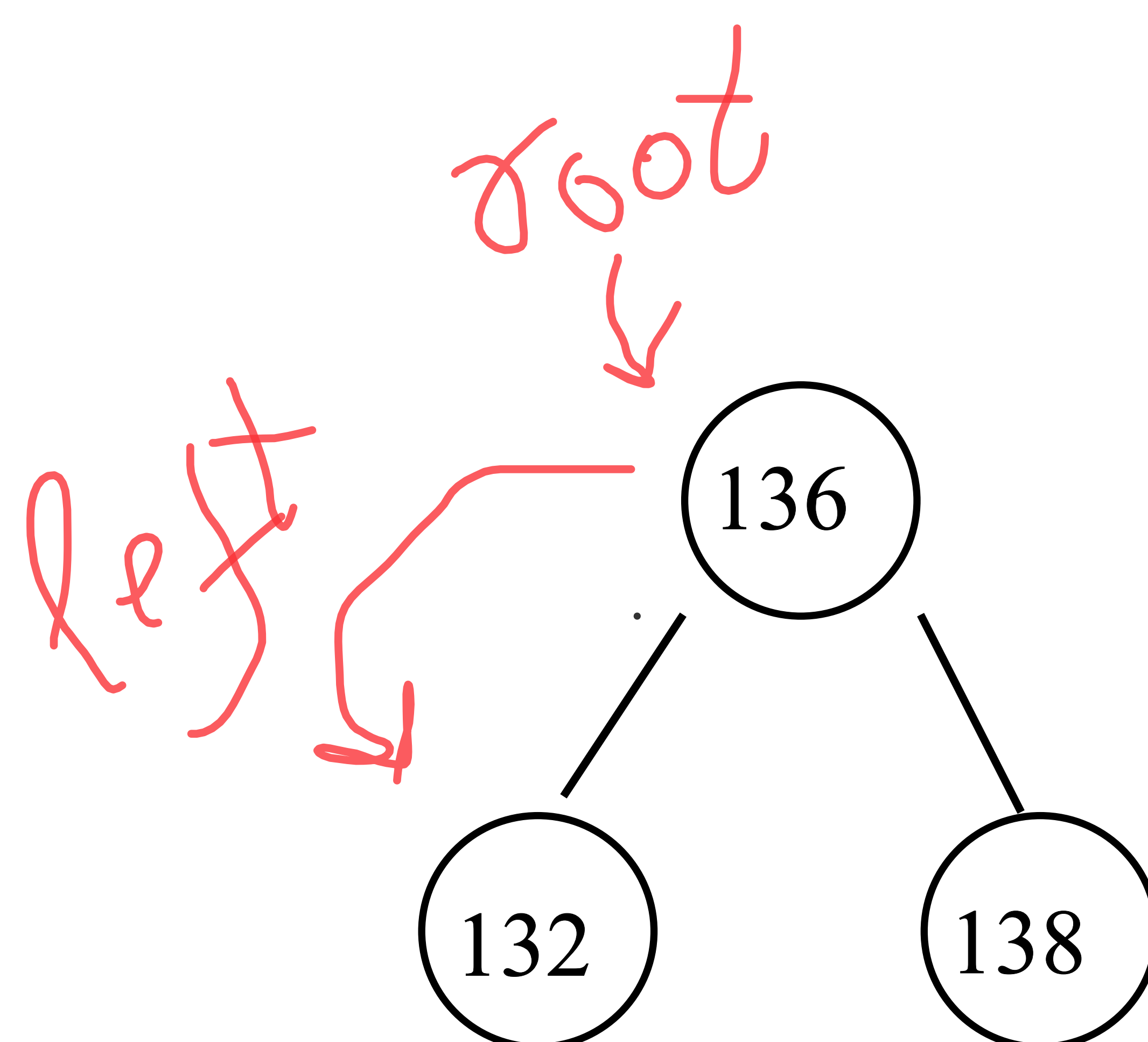
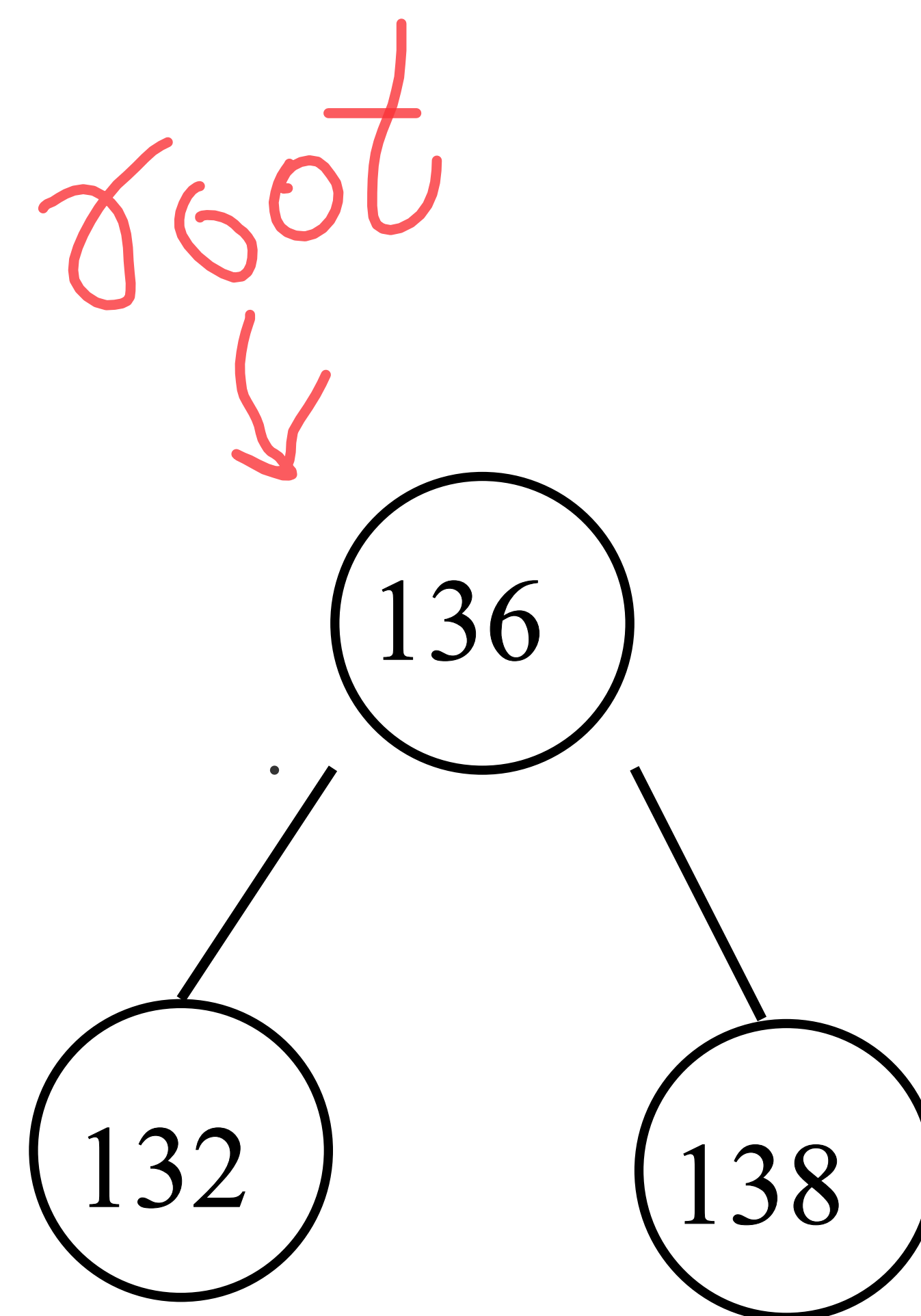
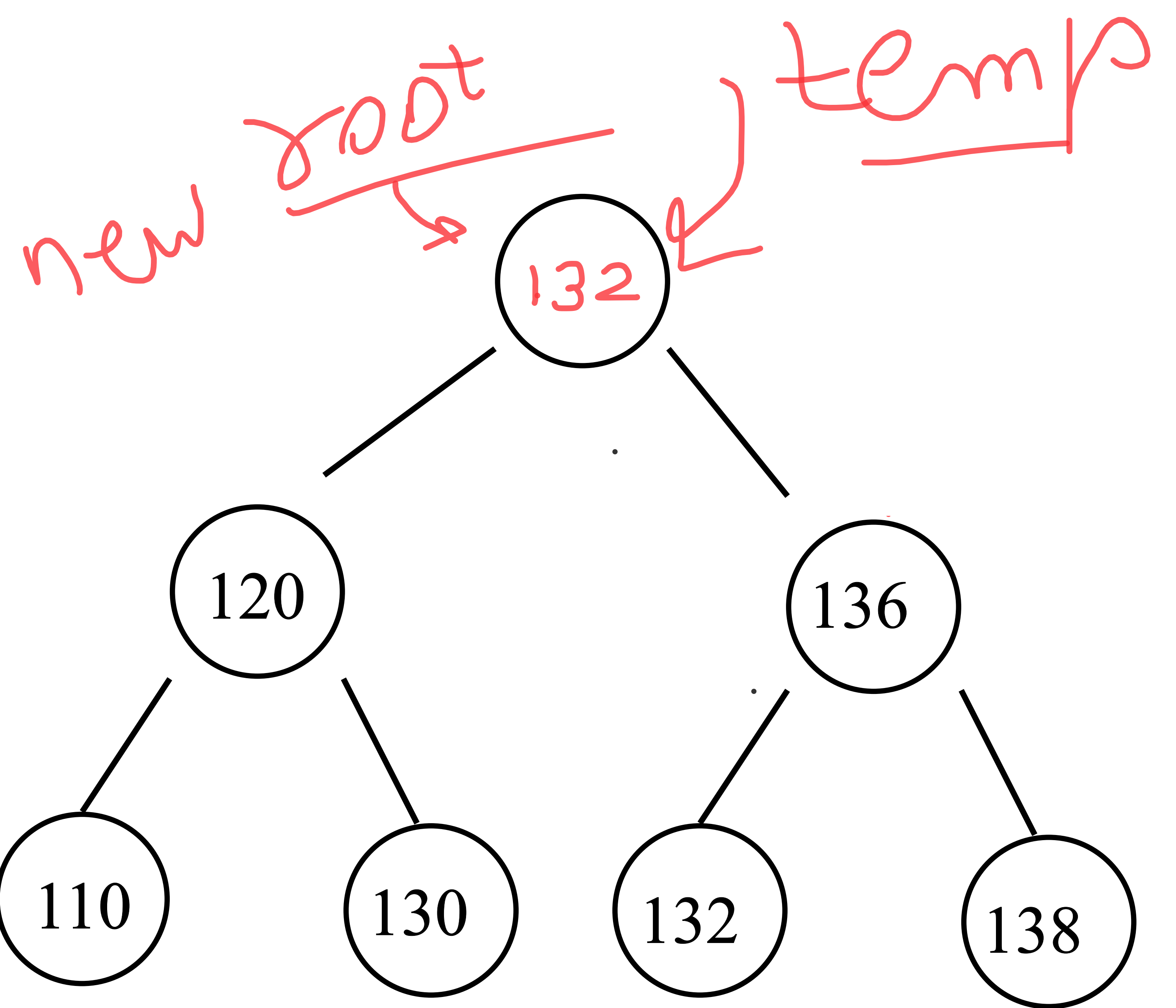
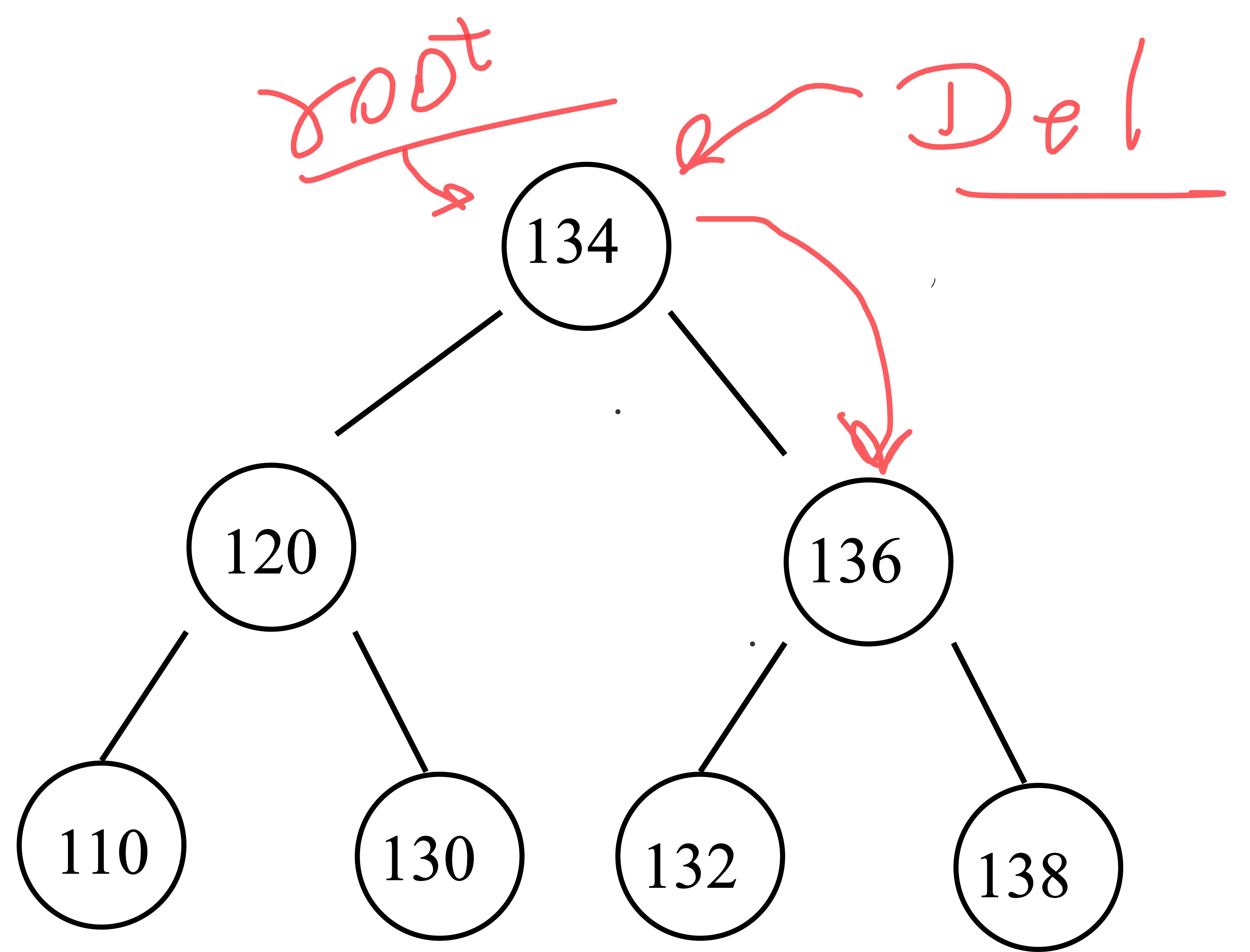
136 → 136 → 132
↓ ↓ ↓
root Val
Recurse

del called
val smaller than the root 132 > 136

again del called at line 199

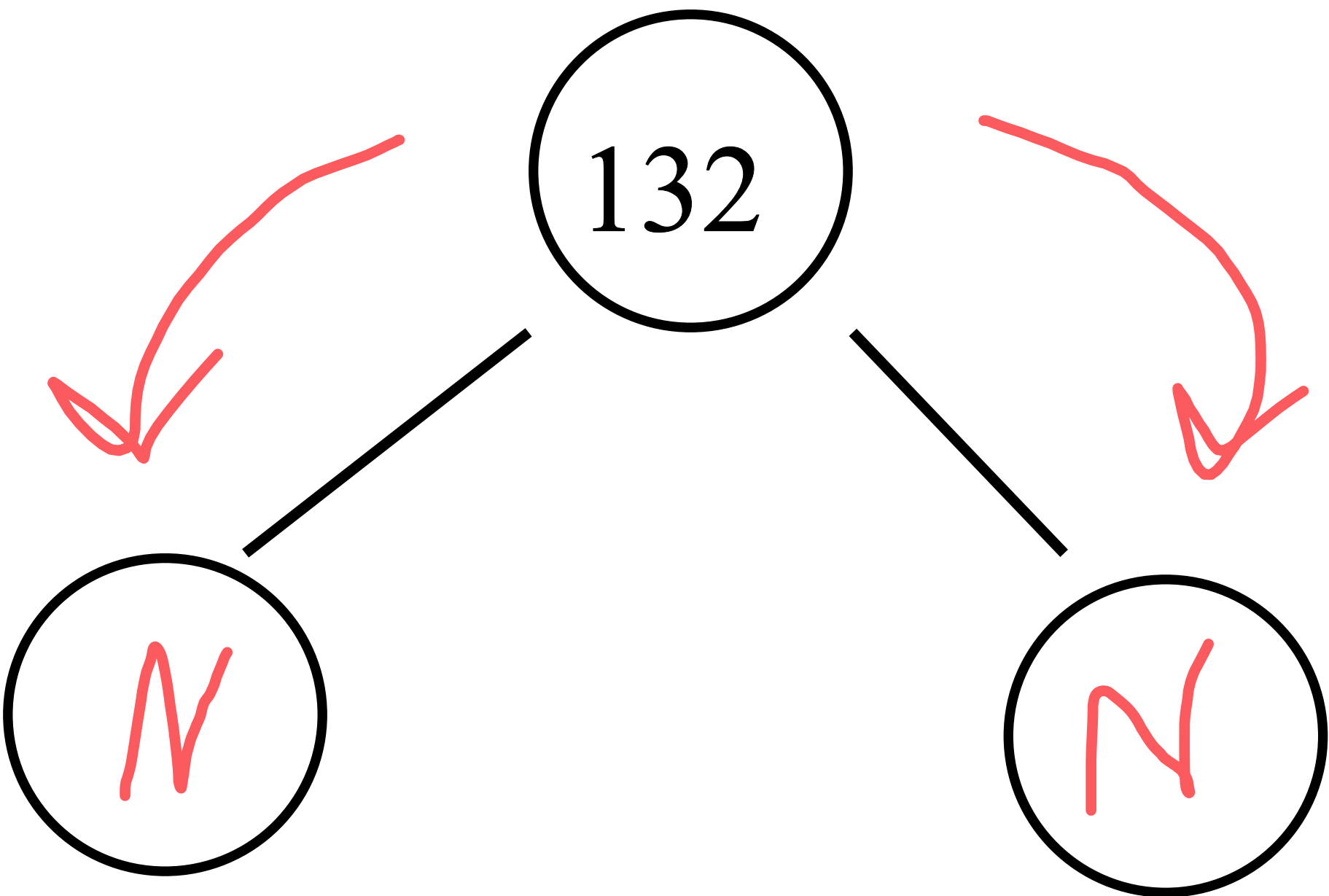
`root->left = Delete (root -> left , value);`

132 ↓ 132 ↓ 132

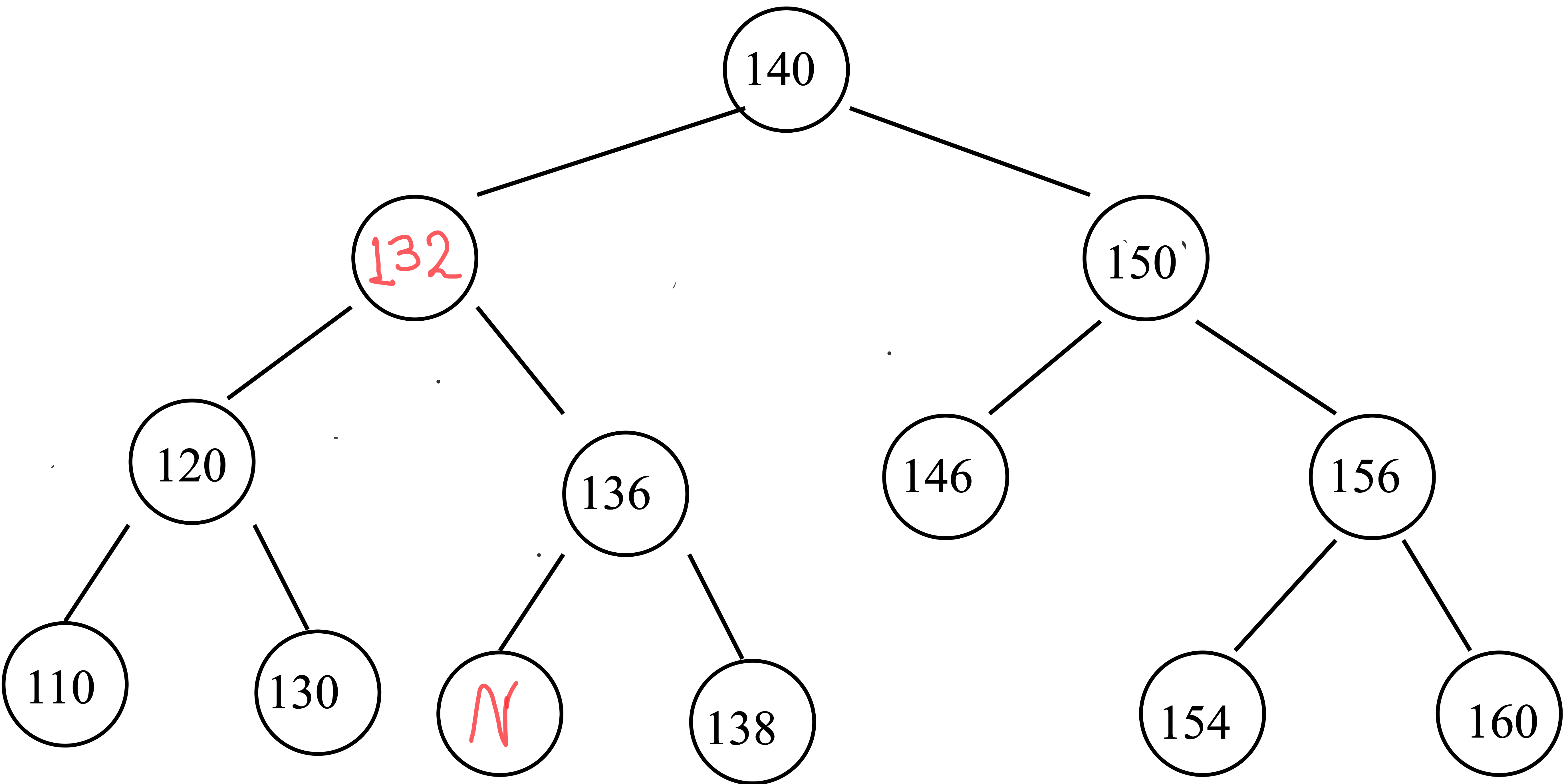
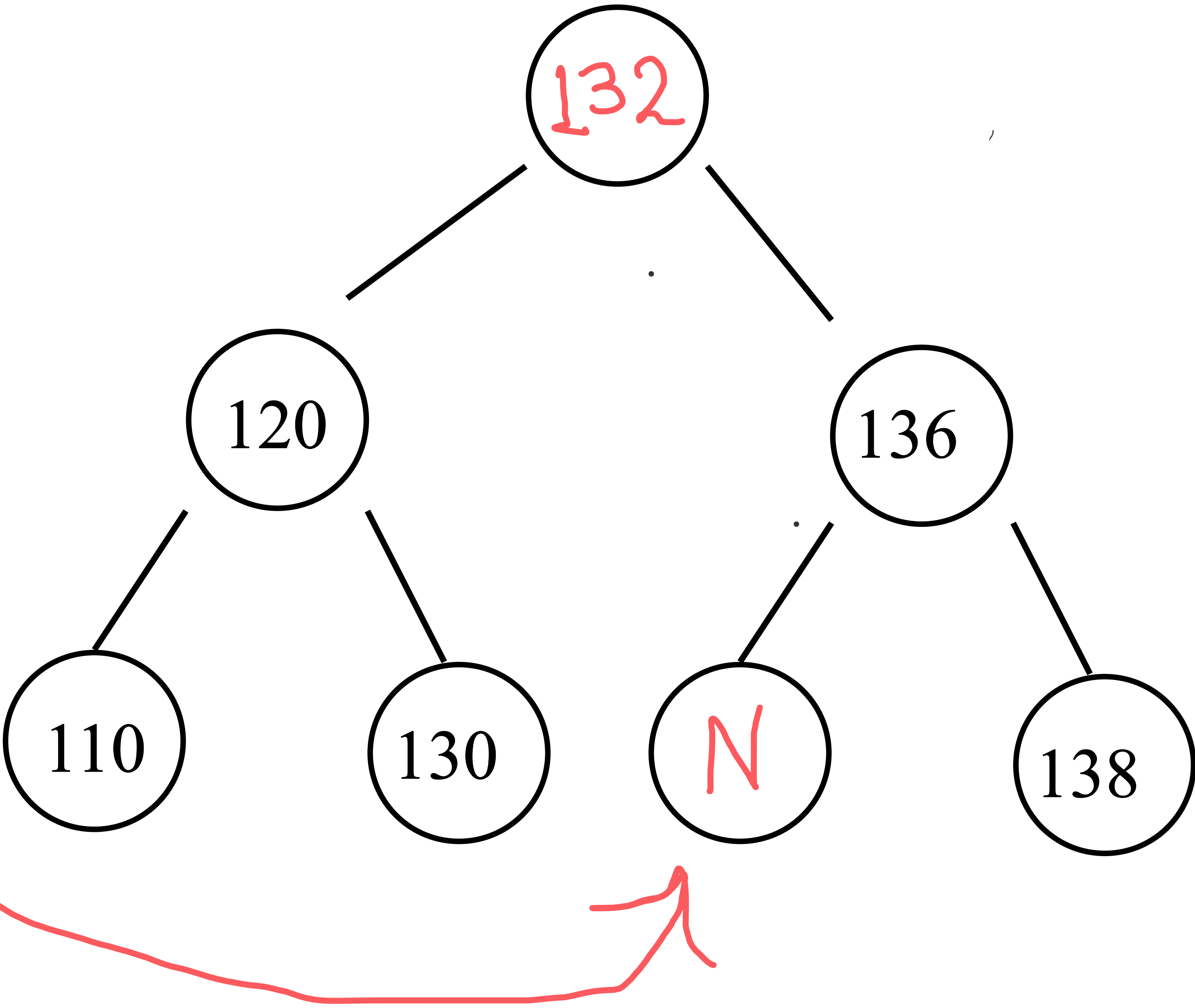


Enters the Del fun and heads straight for the cond 4a

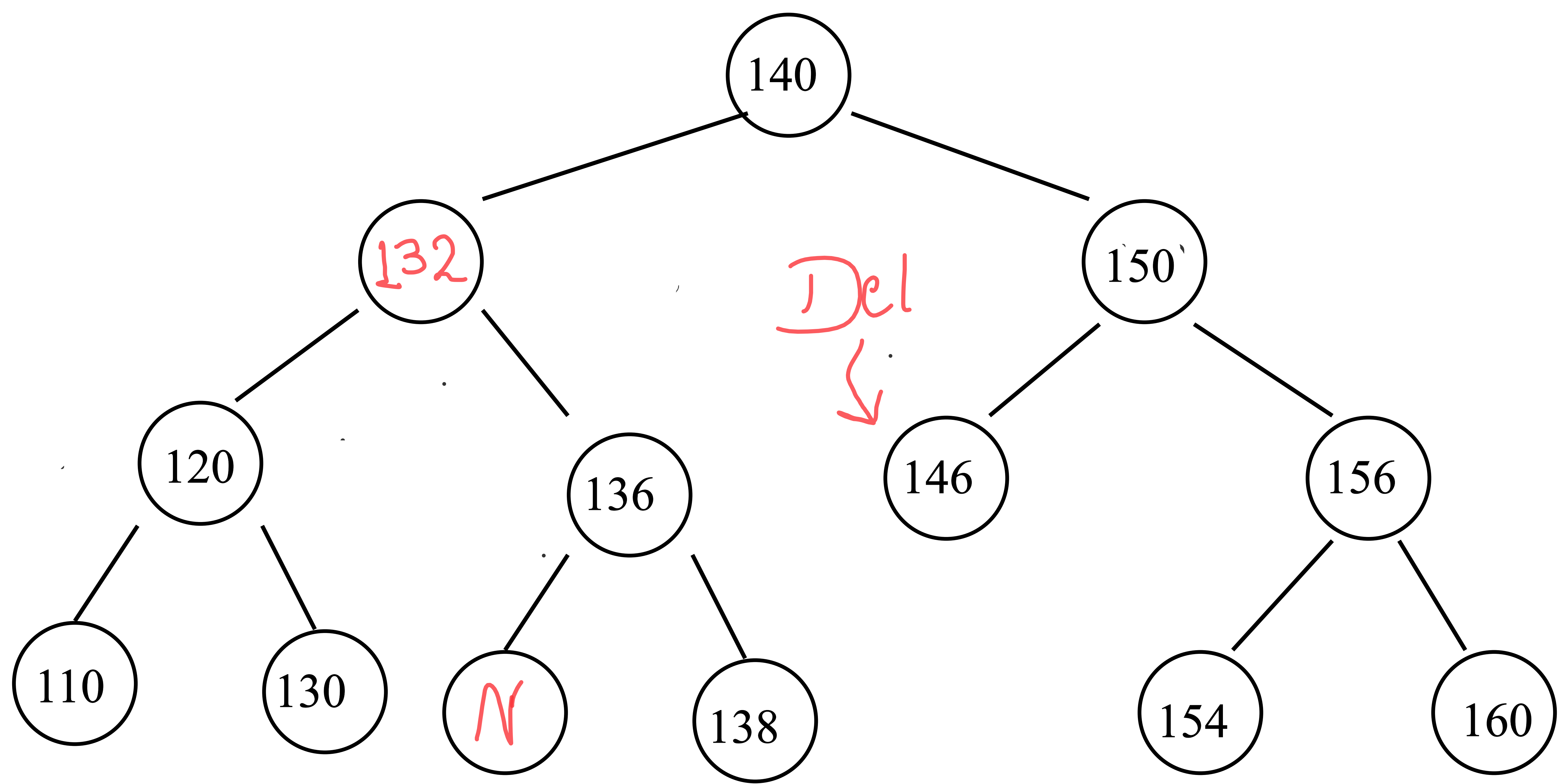
where both left and right of the node 132 are NULL



So the node 132 get deleted and NULL is returned



cond :4d wrt AVL tree for Balancing

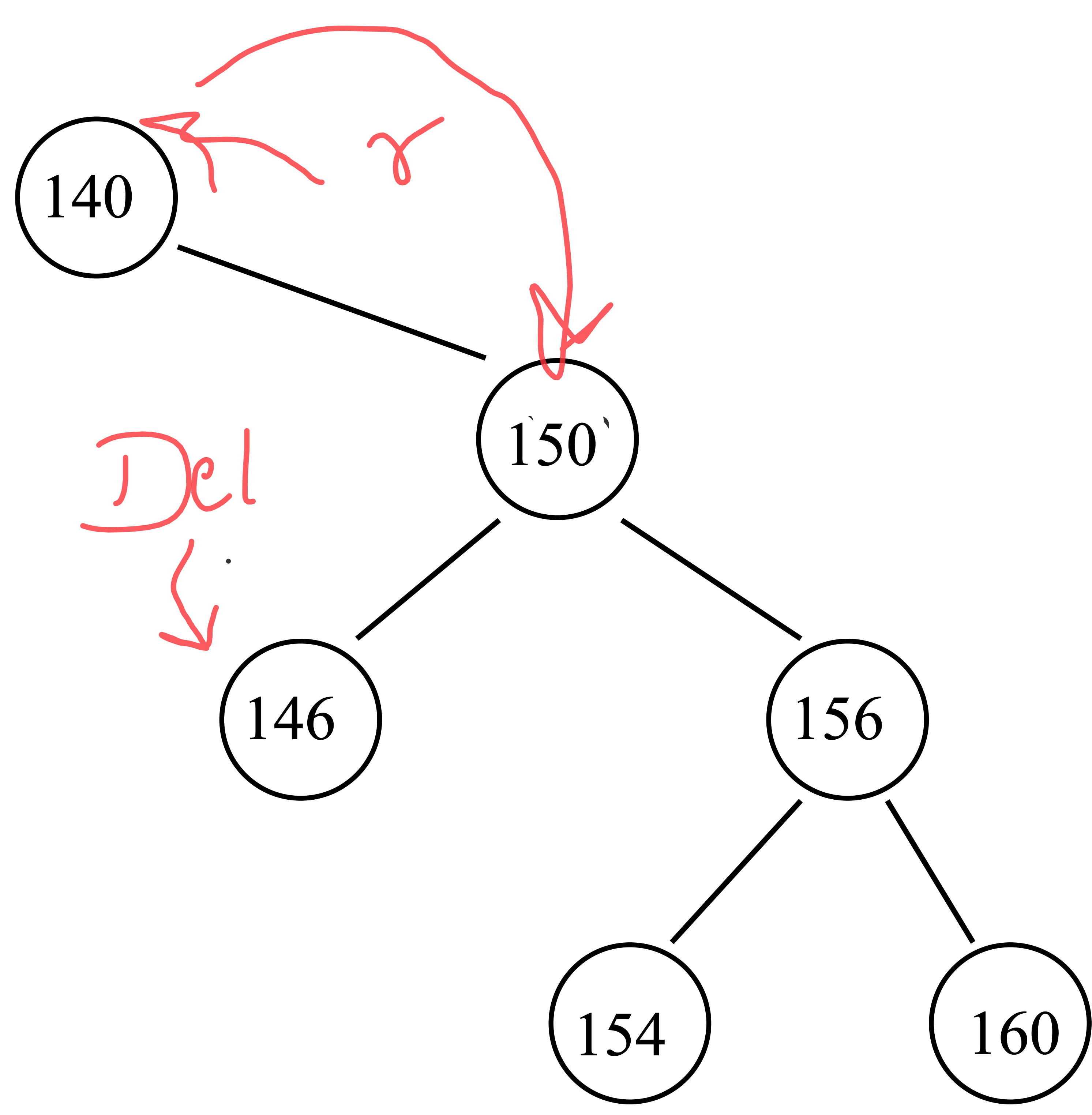


from the main the Del(140 , 146)

↑ root ↑ val

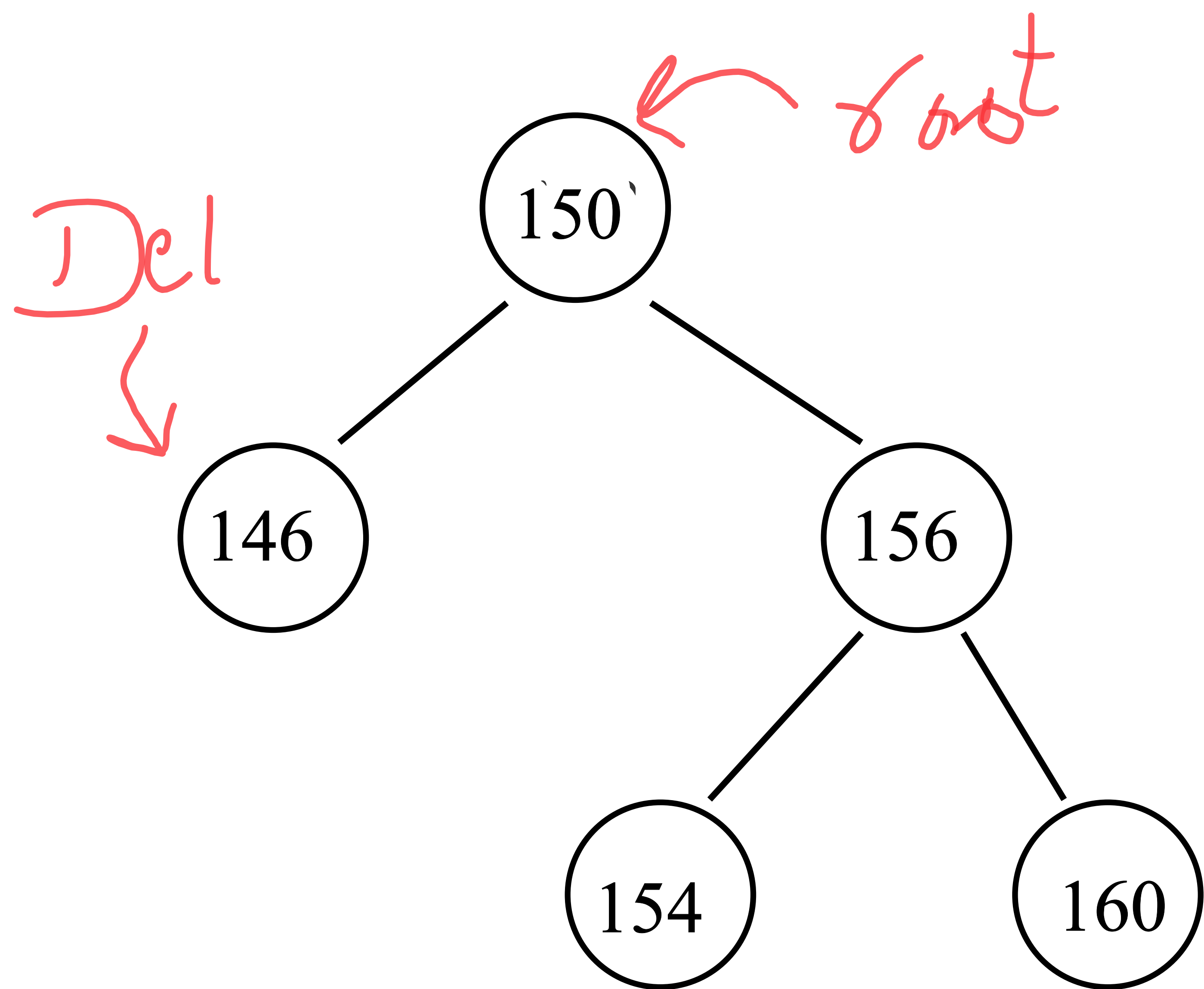
146 > 140 True greater

● root->right=Delete(root->right , value);
150 150 146

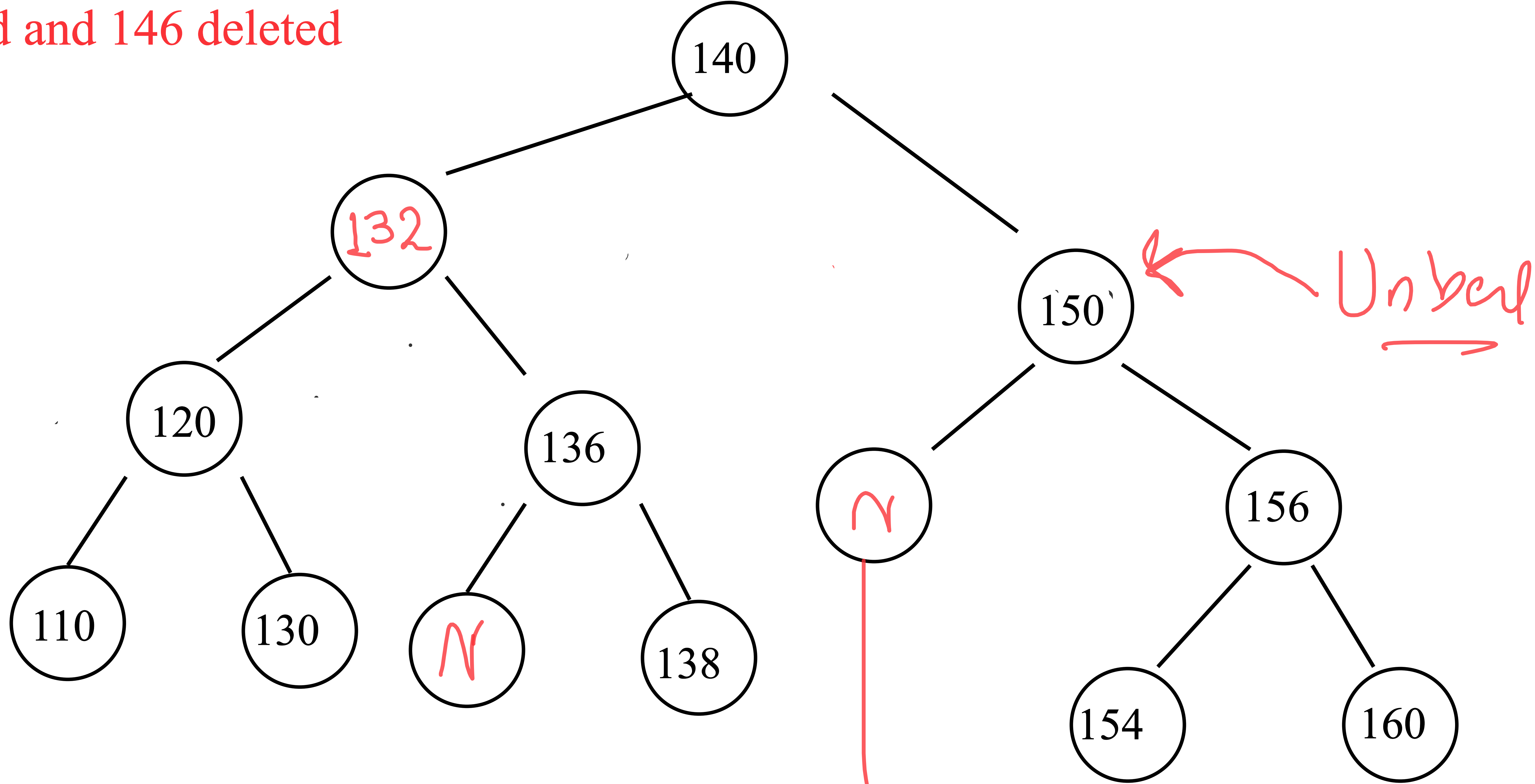


146 < 150 true lesser

● root->left=Delete(root->left,value);
146 146 146



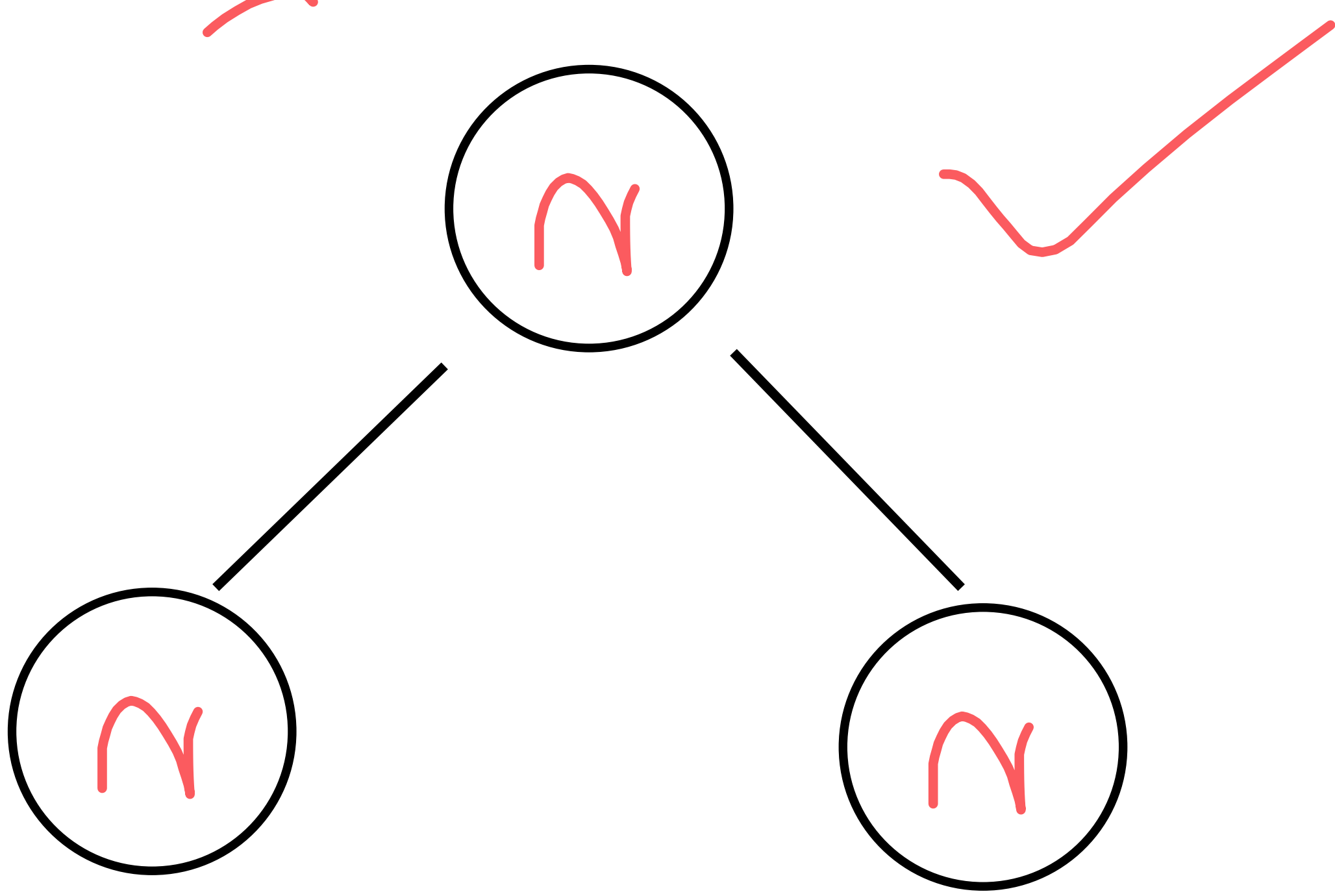
Del called and 146 deleted



● root->left=Delete(root->left,value);

146 = NULL

Checks the balance of Node which is just set to NULL



Balanced

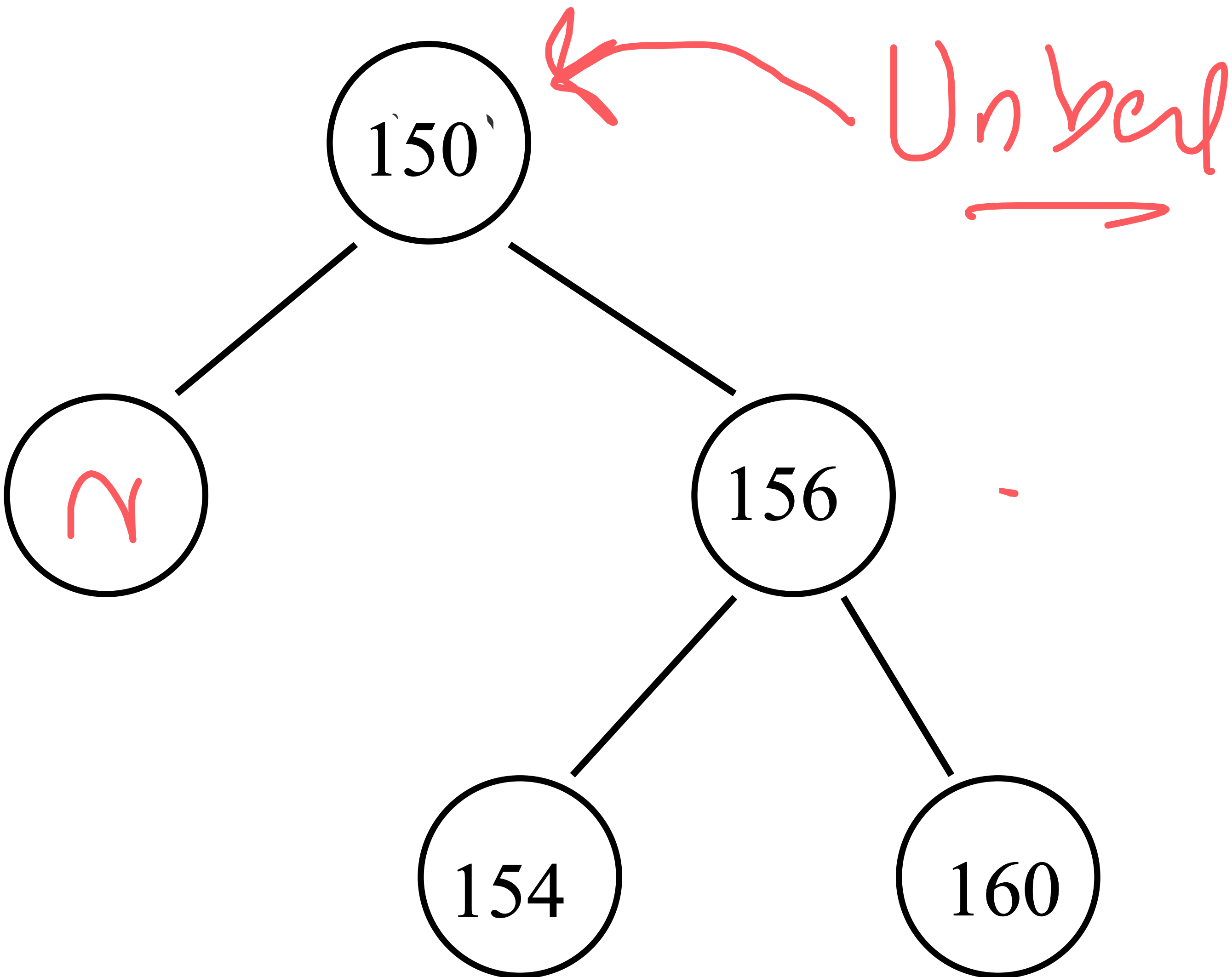
then after that it moves to line 176

● root->right=Delete(root->right , value);

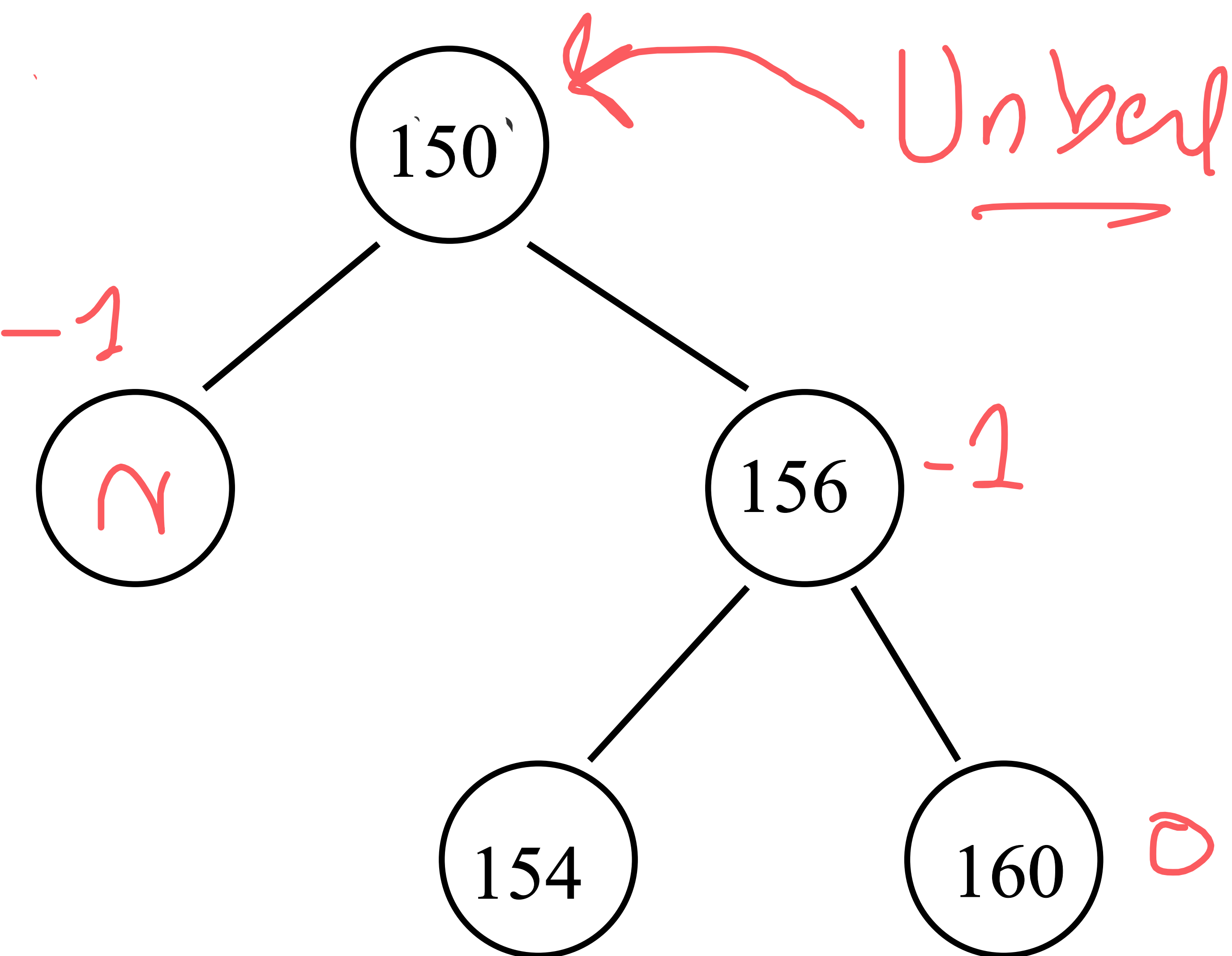
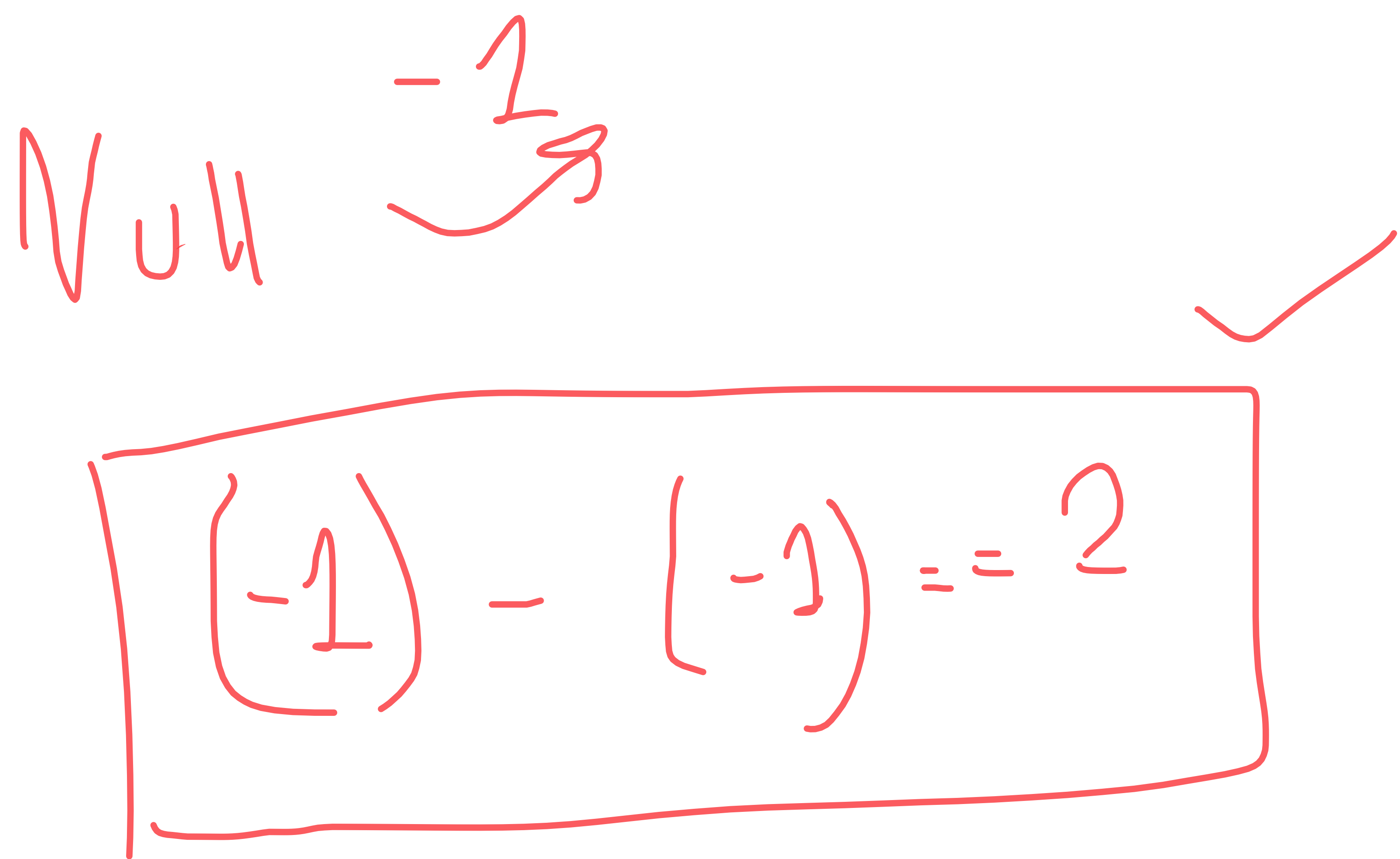
150 = 150 146

then futher checks if there is any imbalance at line 182

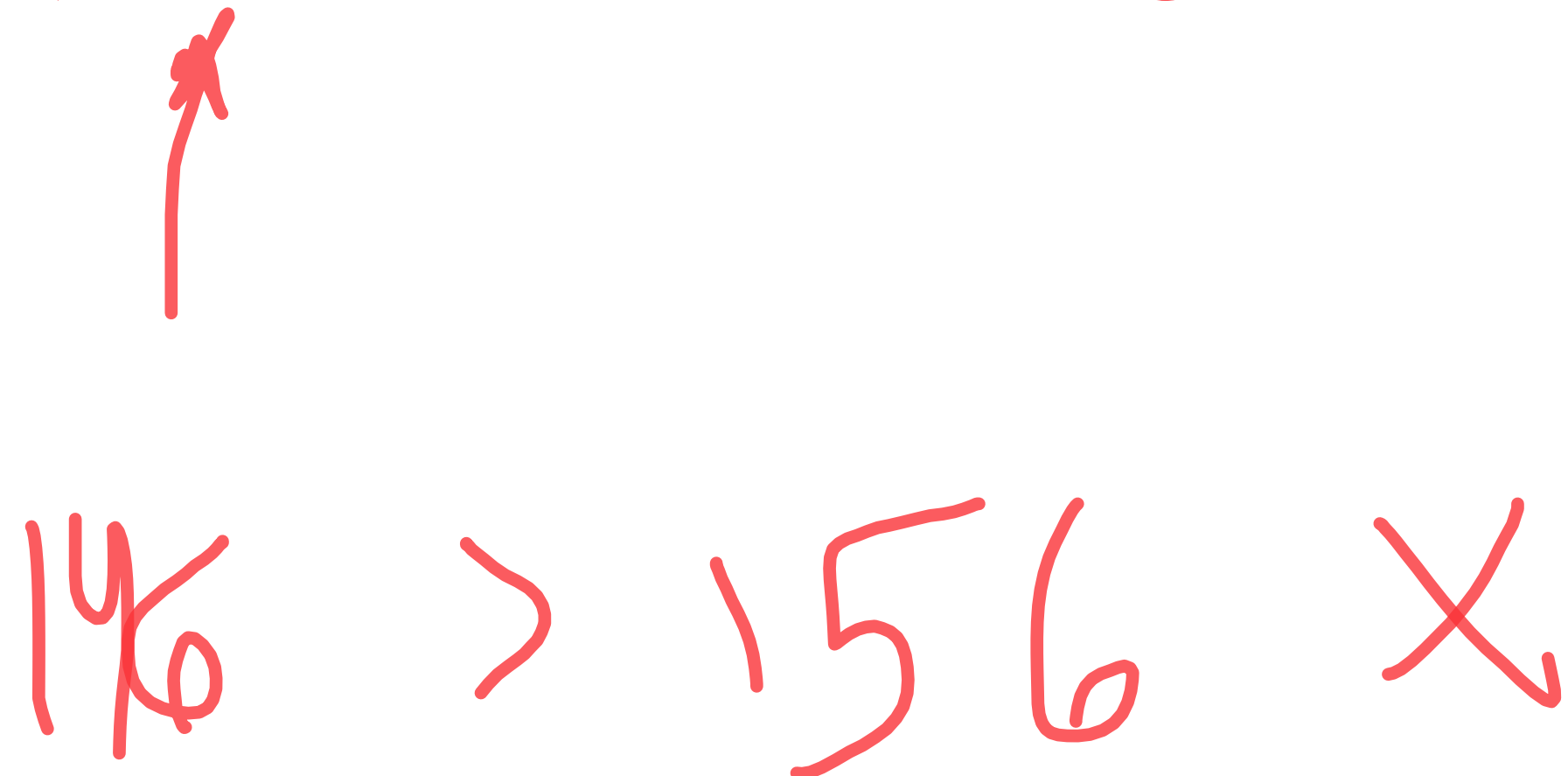
There is imbalance caused



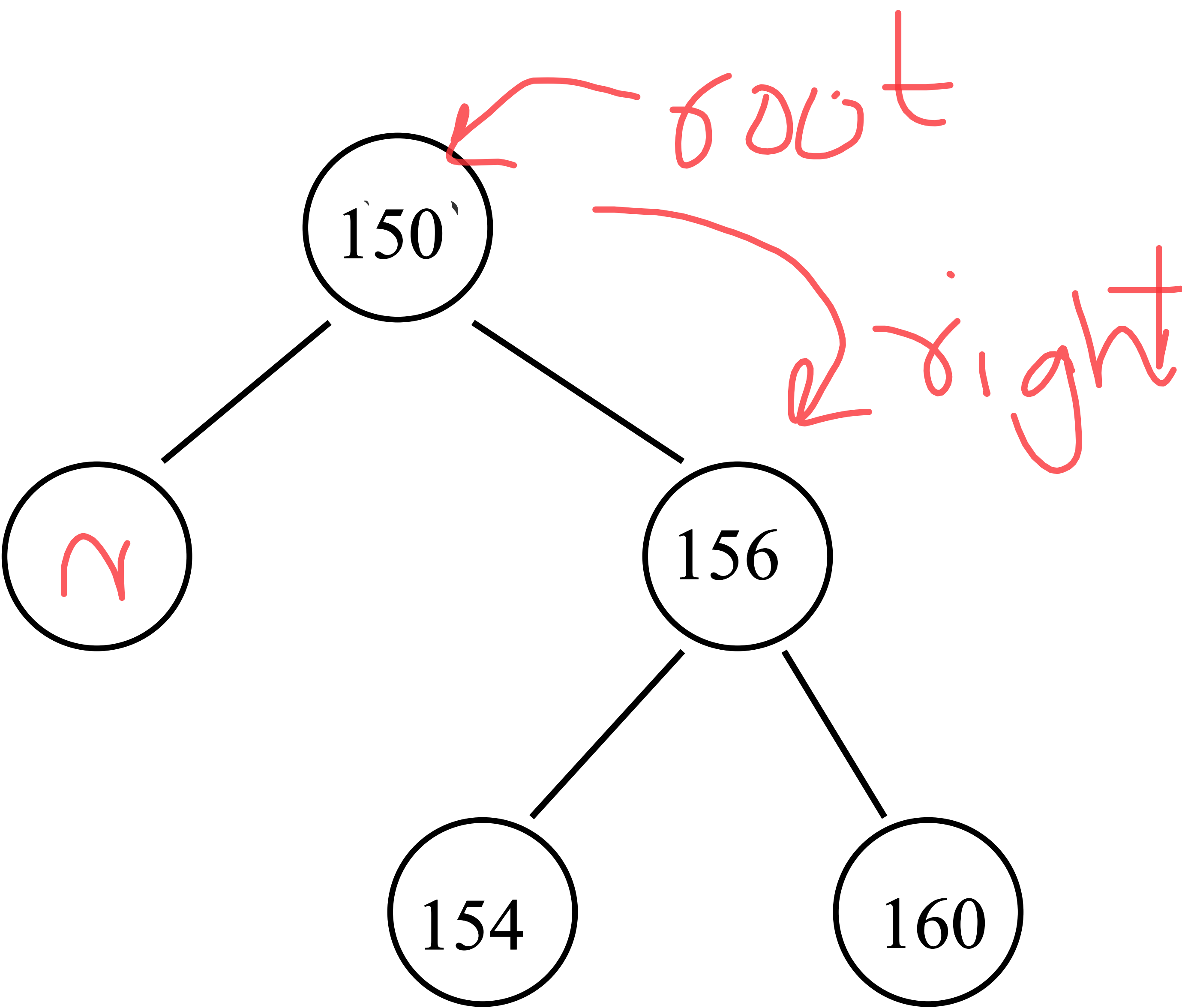

```
if ((getHeight(root->right) - getHeight(root->left)) == 2)
```

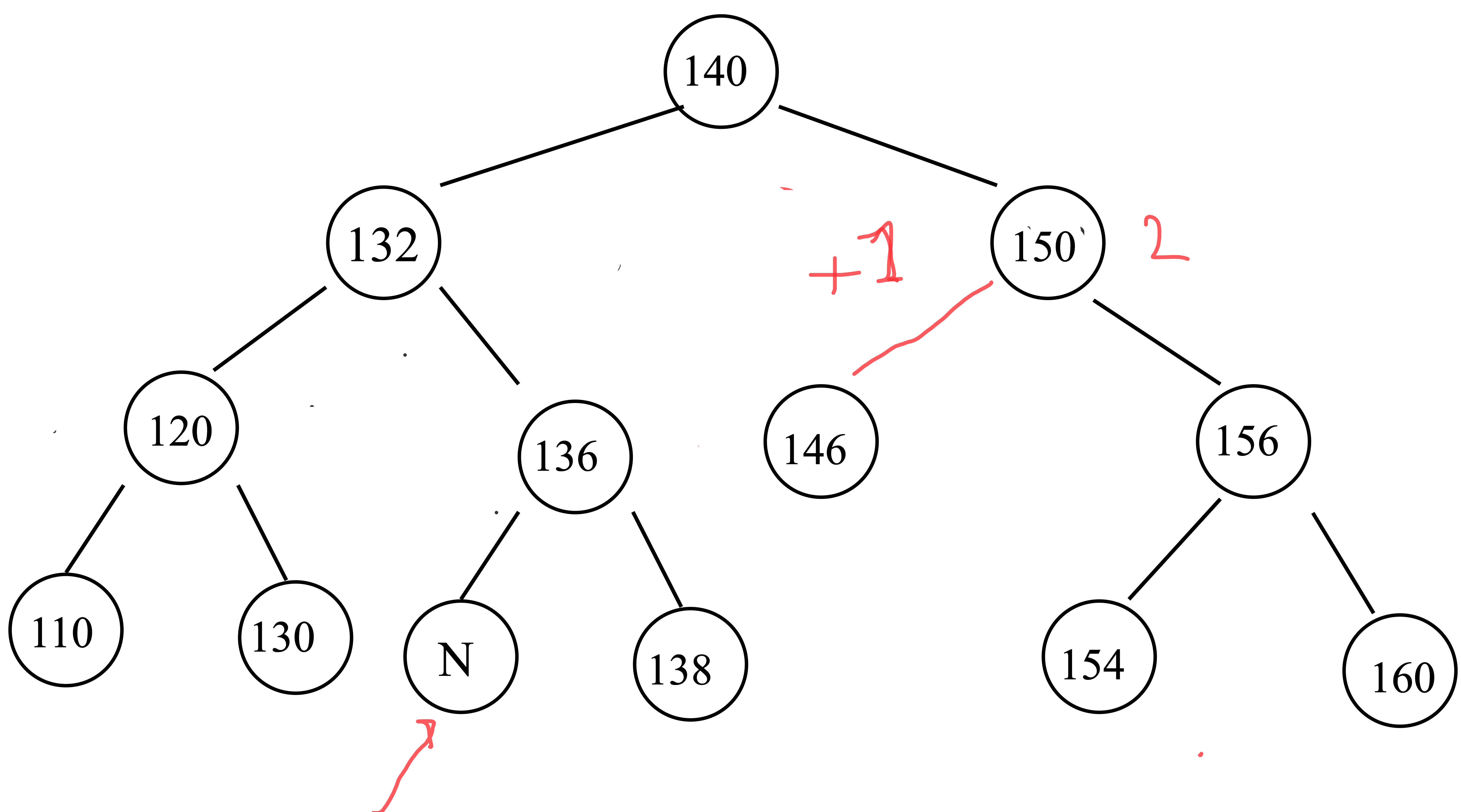
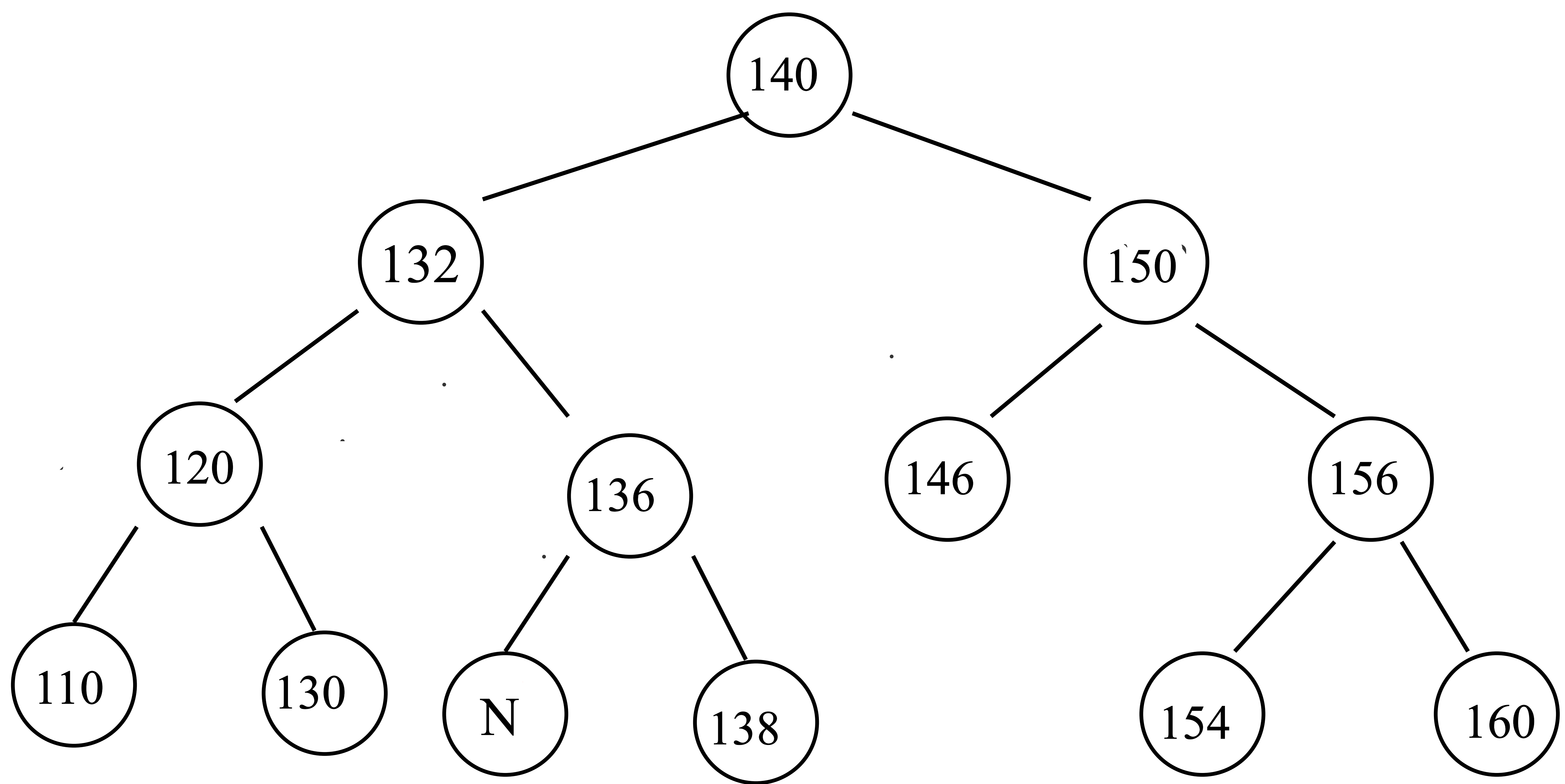


```
if (value > root->right->data)
```

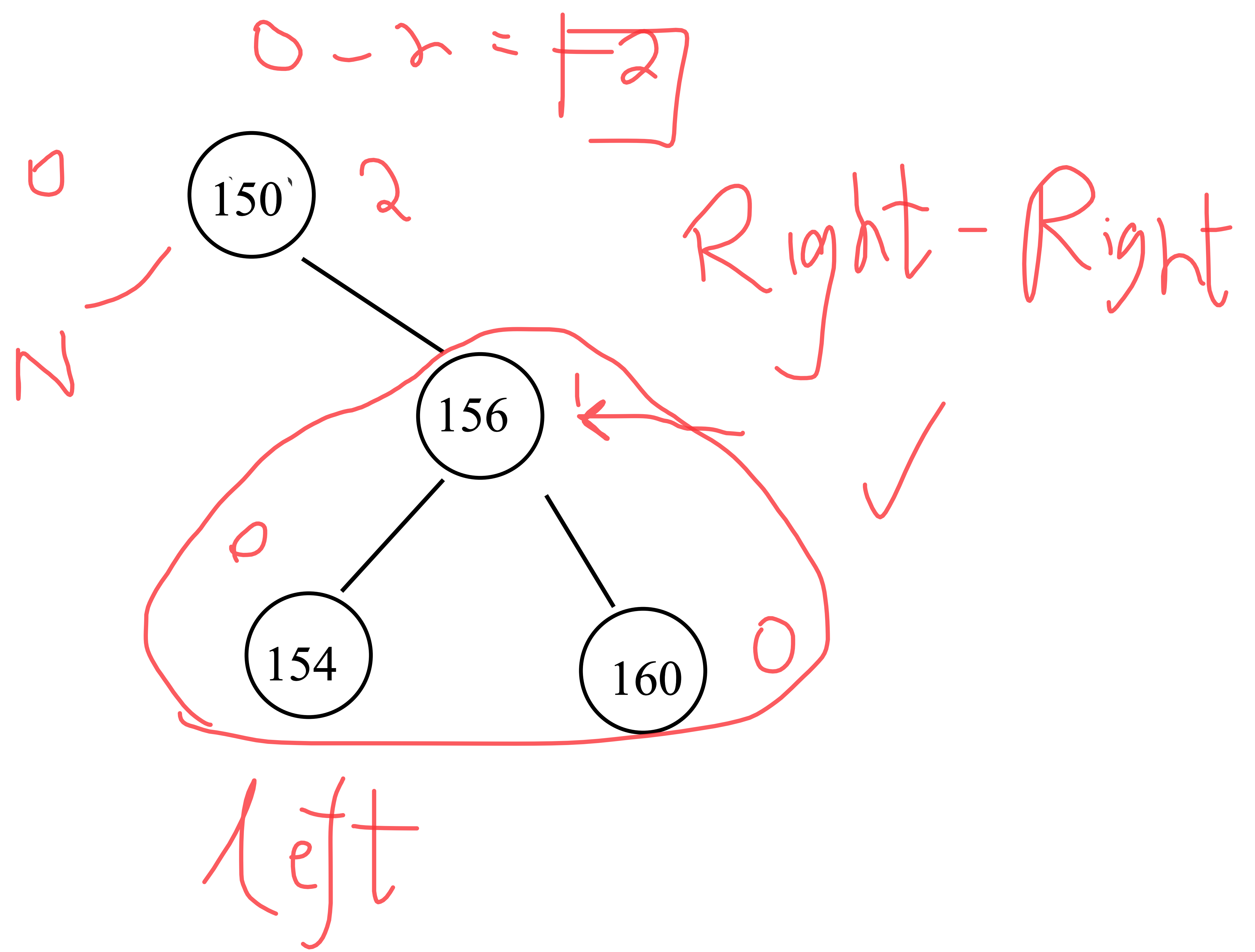


```
else
```





$$+1 - (-1) = 2$$



After Balance

